AllSaints Music: Offering personalized music services across regions with a streamlined data pipeline on cloud

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About AllSaints Music

Founded in 2020, AllSaints Music is a tech startup specializing in digital video and audio services. Partnering with leading hardware manufacturers like Oppo, Vivo and Xiaomi, as well as music producers and entertainment platforms in different countries, AllSaints Music offers a wide range of audio content and personalized music services. Its music streaming application currently has more than 50 million users around the world.

Industries: Technology

personalized mu
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around the world

Location: China

Storage (https://cloud.google.com/storage/),

Google Cloud (https://cloud.google.com/)

BigQuery (https://cloud.google.com/bigquery/)

Cloud Storage (https://cloud.google.com/storage/)

Firebase (https://firebase.google.com/)

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Google Cloud results

- Guarantees data compliance in different countries under one unified data transferring and storage structure
- Helps create more personalized music recommendations for users through more detailed data collection enabled by Firebase
- Supports exponential growth of ad revenues with data analytics

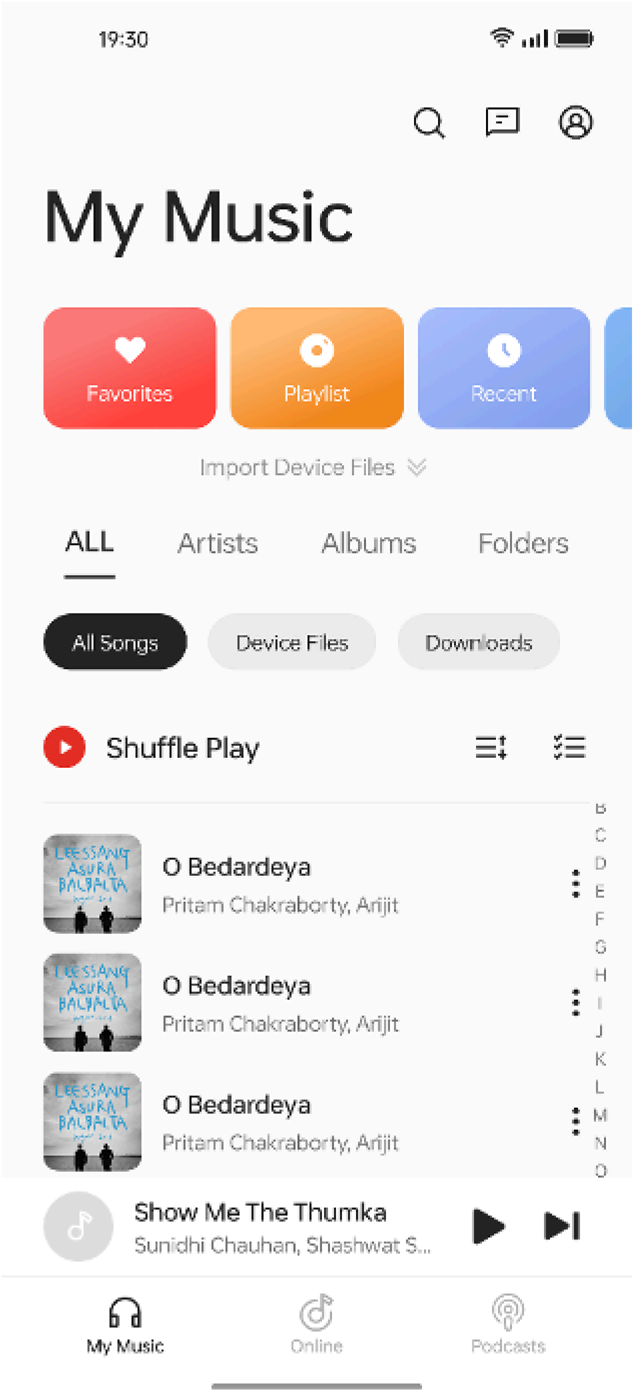
40% higher efficiency in cross-region data collection

results generated
by BigQuery

Audio content is central to many people’s lives. Although the internet provides easy access to all genres of audio content from music to podcasts, discovering what you like or aggregating your preferred audio clips in a sea of content can be challenging.

AllSaints Music seeks to address this gap by creating seamless and personalized audio streaming experiences. Founded in 2020, the tech startup partners with hardware manufacturers like Oppo, Vivo, and Xiaomi to launch its music streaming application (https://play.google.com/store/apps/details?id=com.heytap.music) on mobile devices. Through collaborating with music producers and entertainment platforms in different countries, AllSaints Music has built an extensive audio content collection that caters to all types of music tastes and listening habits. The application currently has more than 50 million users around the world, mostly in China, Southeast Asia and India.

"Our goal is to offer an audio streaming platform where everyone can find the content they like no matter what their music taste is or what language they speak, and that’s why we have an ever-expanding content collection," says Zhengyuan Wei, project and product manager at AllSaints Music. "Based on our immense content library, we’ve been striving to offer personalized features like custom playlists to help users access their favorite music more easily and constantly discover new content that they would like."



To strengthen its personalized offering, AllSaints Music relies on data analytics to understand its users’ preferences and deliver better recommendations. As a data-driven company, it also continuously uses data insights to improve its business and marketing strategies.

At first, the tech startup deployed its data pipeline on a public cloud, which offered modest coverage with a limited number of data centers across regions. The AllSaints Music team therefore could only store all its user data in Singapore and needed to manually configure its data storage to comply with data regulations in different countries. This was not only labor-consuming, but also resulted in inaccurate data collection and frequent data loss, because data had to be migrated through long distance and between different database models.

To enhance the efficiency of its analytics workloads while ensuring data compliance across countries, AllSaints Music needed a global cloud platform that supports high-performance cross-region data collection, analytics, and storage. In early 2022, the tech startup decided to migrate its data pipeline to Google Cloud (https://cloud.google.com/), because the latter has widely distributed data centers worldwide that are in compliance with local regulations and a full set of data tools.

"To optimize our personalized services and data-driven strategies across multiple regions, we needed a powerful, fully compliant global platform to support our data analytics workloads," explains Wei. "Google Cloud perfectly fulfills our requirements for data protection and processing efficiency."

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Ensuring data security and privacy across regions

With the training sessions and technical support provided by the Google Cloud team, AllSaints Music smoothly migrated its data pipeline to Google Cloud within two months, with all its data stored in Cloud Storage (https://cloud.google.com/storage#section-1) across zones and regions. With the data security and privacy measures of Google Cloud, AllSaints Music can ensure that its operations remain compliant with local regulations

"Data security and privacy is the priority in our data analytics workloads," notes Wei. "Since data regulations differ in each country, we needed to configure various data pipelines and storage for adhering to compliance, which complicated our operations. With Google Cloud, we can rest assured that all our data is in line with data compliance."

Plus, Allsaint Music’s data is no longer being migrated through the multiple, disparate data center networks of its previous public cloud provider, since Google Cloud is built upon a single network. This has prevented any network failures and data loss since the migration. Besides, the pay-as-you-go pricing of Cloud Storage also enables the tech startup to be more cost-effective, since it only pays for the storage it uses.

Offering truly personalized features with Firebase

To fully understand its users' preferences and music listening habits, AllSaints Music leverages Firebase (https://firebase.google.com/) to collect data related to user portraits and behavior. Before the migration to Google Cloud, its data team had to create several data collection templates for users in different countries and manually align data formats. Through Firebase, AllSaints Music can build a unified data collection system, which not only automatically aggregates user data like location and device, but also lets its team configure the collection parameters for user behavior data, such as clickthrough rate and listening duration with ease. Therefore, its data collection efficiency is enhanced by 40%.

With more detailed user behavior data at hand, AllSaints Music can now offer personalized features that better fit its users' preferences, which is verified through A/B testing run by Firebase. For example, the music recommendation feature on its app now receives more positive feedback, with users clicking on the recommended songs more frequently and listening to them for longer periods.

"Firebase has helped us significantly improve our personalized features. With its highly automated and powerful data collection capabilities, we're able to effortlessly gather all the data we need to understand what our users like and offer tailored music streaming experiences," says Wei.

Enhancing ad performance with data insights from BigQuery

AllSaints Music also imports all the data collected by Firebase into BigQuery (https://cloud.google.com/bigquery) for data analytics to gain business insights. Since Google Ads (https://ads.google.com/intl/en_us/home/) can be seamlessly integrated with Firebase, AllSaints Music is able to analyze its ad data on BigQuery more easily. Before the migration, the company used data tools from several providers to build its data pipeline, and its data team had to switch between different platforms to retrieve data and insights. With the whole data pipeline supported by Google Cloud, the team can rely on one dashboard to transfer and analyze its data, which results in 35% higher productivity and more useful data insights.

"By leveraging BigQuery to analyze data more efficiently, we have gained more insights to improve our advertising and business strategies," Wei adds. "Both our ad clickthrough rate and revenue have grown exponentially since the migration to Google Cloud."

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Unlocking the value of data to create new features

AllSaints Music is always looking to expand the features of its app to offer more comprehensive and diverse entertainment services. It is currently using artificial intelligence (AI) technology to upgrade features like music composition, karaoke, and live broadcasting. To improve its AI models, AllSaints Music will tap on Pub/Sub (https://cloud.google.com/pubsub) by Google Cloud, which exports data from its app to be analyzed for insights. It also plans to adopt Looker Studio (https://cloud.google.com/looker-studio) for data visualization, so that its team can handle all the data workloads on Google Cloud more efficiently.

Wei says, "Deploying our cross-region data pipeline on Google Cloud has enabled us to greatly enhance our personalized services and business results by processing data more securely and efficiently. We’ll continue to unlock the value of our data with the excellent analytics capabilities of Google Cloud."

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